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ORIGINAL PAPER

Persistent olfactory dysfunction as an indicator of cognitive impairment in long COVID-19 syndrome: implications for monitoring and rehabilitation

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Abstract

Background Persistent olfactory dysfunction (OD) and cognitive impairment are among the most frequently reported sequelae of long term infection with SARS-CoV-2 (long COVID-19). However, the association between these conditions remains unclear. This study investigated the correlation between OD and cognitive impairment in patients recovering from COVID-19 to identify the implications for therapeutic and rehabilitation strategies.

Materials and methods A cross-sectional study of adult patients diagnosed with long COVID was conducted at a healthcare centre in Brazil. Olfactory function was assessed using the Connecticut Chemosensory Clinical Research Centre (CCCRC) test, and cognitive performance was evaluated using the Montreal Cognitive Assessment (MoCA). Statistical analyses included odds ratios (OR) and linear regression to explore the association between OD severity and cognitive scores, adjusting for potential confounders such as age, sex, and comorbidities.

Results A total of 241 patients (age: 48.60 ± 12.68 years; 73% female) were included. Cognitive impairment (MoCA < 23) was present in 64% of the participants. OD was identified in 92% of the patients and ranged from mild to anosmia. Linear regression analysis showed a weak yet statistically significant correlation between the CCCRC and MoCA scores ($R = 0.14$, $p = 0.02$). An OR of 2.87 (95% CI: 1.57–5.25, $p = 0.00$) indicated higher odds of cognitive impairment in patients with severe OD.

Discussion This study supports the hypothesis that there is a weak yet significant association between OD and cognitive impairment in patients with long COVID. These findings underscore the importance of early screening for olfactory dysfunction as a potential marker of cognitive monitoring and the need for intervention in this population.

Keywords COVID-19 · Long COVID · Olfactory dysfunction · Cognitive impairment

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Introduction

Olfactory dysfunction (OD) is a characteristic and highly prevalent symptom among patients in the acute phase of Coronavirus Disease of 2019 (COVID-19), as well as among those who develop long COVID [1–3]. Long COVID-19 syndrome is defined as the presence, persistence, or development of signs, symptoms, and/or conditions 4 weeks or more after the initial infection with SARS-CoV-2, which cannot be attributed to an alternative diagnosis [4, 5].

Worldwide, an estimated 15 million people may be affected by persistent OD due to the COVID-19 pandemic, representing a significant global public health issue [6]. The potential causes of prolonged OD after COVID-19 include damage to basal cells, ongoing inflammation, and chronic SARS-CoV-2 infection of the olfactory epithelium [7, 8]. Chronic inflammation may modulate gene expression and alter the function of olfactory epithelial basal cells, including neural regeneration, inflammatory signalling, and immune cell proliferation [9].

Chronic olfactory loss negatively affects daily activities and social participation, such as food preparation and consumption, personal hygiene, and diminishes the ability to detect harmful stimuli such as gas, smoke, and fire [10–12]. It is also associated with an increased risk of mood disorders, including anxiety and depression [13–15], as well as neurodegenerative diseases such as Parkinson disease and Alzheimer disease [16–19].

A similar pattern of persistent OD and early damage to the olfactory and limbic regions of the brain in patients with COVID-19 is observed in the early stages of Alzheimer disease, Parkinson disease, and Lewy body dementia [20]. Differentiating long COVID from dementia is difficult due to the overlap of other signs of neurodegeneration such as cognitive and emotional disturbances associated with COVID-19 [20–22].

The potential association between olfactory and cognitive impairments in patients with COVID-19 remains an ongoing debate. While some studies found no significant correlation between these 2 deficits [23–26], others have reported significant associations of varying magnitudes [27–39]. The reasons for these discrepancies are unclear; however, variations in the cognitive function assessment and clinical characteristics of the patients, including the time elapsed since the primary COVID-19 infection, may play a role.

Data on the association between olfactory function and cognition in patients with long COVID remain scarce in the literature. This study aimed to investigate possible associations between chronic OD and cognitive impairment in individuals with long COVID.

Materials and methods

Population and study design

This cross-sectional study was conducted between 9 September 2020 and 20 October 2023 in the city of Belém, Pará, Brazil. The study population included 440 patients with long COVID taken from a follow-up program for long COVID at a public university, a database of patients with persistent symptoms after COVID-19.

The criteria established by Raveendran [40] were employed for the diagnosis of long COVID, which initially encompassed categories of symptomatic and asymptomatic patients, and subsequently subdivided into confirmed, probable, possible, and doubtful cases based on evidence of previous SARS-CoV-2 infection and the presence of long-term clinical symptoms.

Patients aged 18 years or older with neurological symptoms related to COVID-19 and previously confirmed COVID-19 (evidenced by a positive reverse transcription polymerase chain reaction test) were recruited to participate in the study. Additionally, participants were required to have undergone the Connecticut Chemosensory Clinical Research Center (CCCRC) olfactory test and the Montreal Cognitive Assessment (MoCA) cognitive screening test. Patients with a history of neurological, psychiatric, or cognitive disorders; gustatory or olfactory dysfunction; pregnancy; or an uncertain diagnosis of long COVID were excluded.

This study was approved by the Research Ethics Committee of the State University of Pará (report no. 3,619,141) in accordance with the guidelines established in Resolution 510/16 of the National Health Council (CNS) of Brazil. All participants provided written informed consent.

Flowchart of the study and procedures

Initially, the participants were registered using an electronic form (Google Forms, Google Corporation Inc.), and a medical assessment was scheduled by a neurologist, an otolaryngologist, and a multidisciplinary team. A total of 376 patients 18 years of age or older were assessed in person. Of these, 21 (5%) were excluded because of incomplete sociodemographic data or pre-existing conditions, resulting in a sample size of 355 patients. Among the patients with confirmed long-term neurological symptoms, 241 completed both olfactory and cognitive screening assessments (Fig. 1).

Sociodemographic and clinical data, including educational level, sex, income, and reports of olfactory and cognitive impairment following COVID-19, were collected and

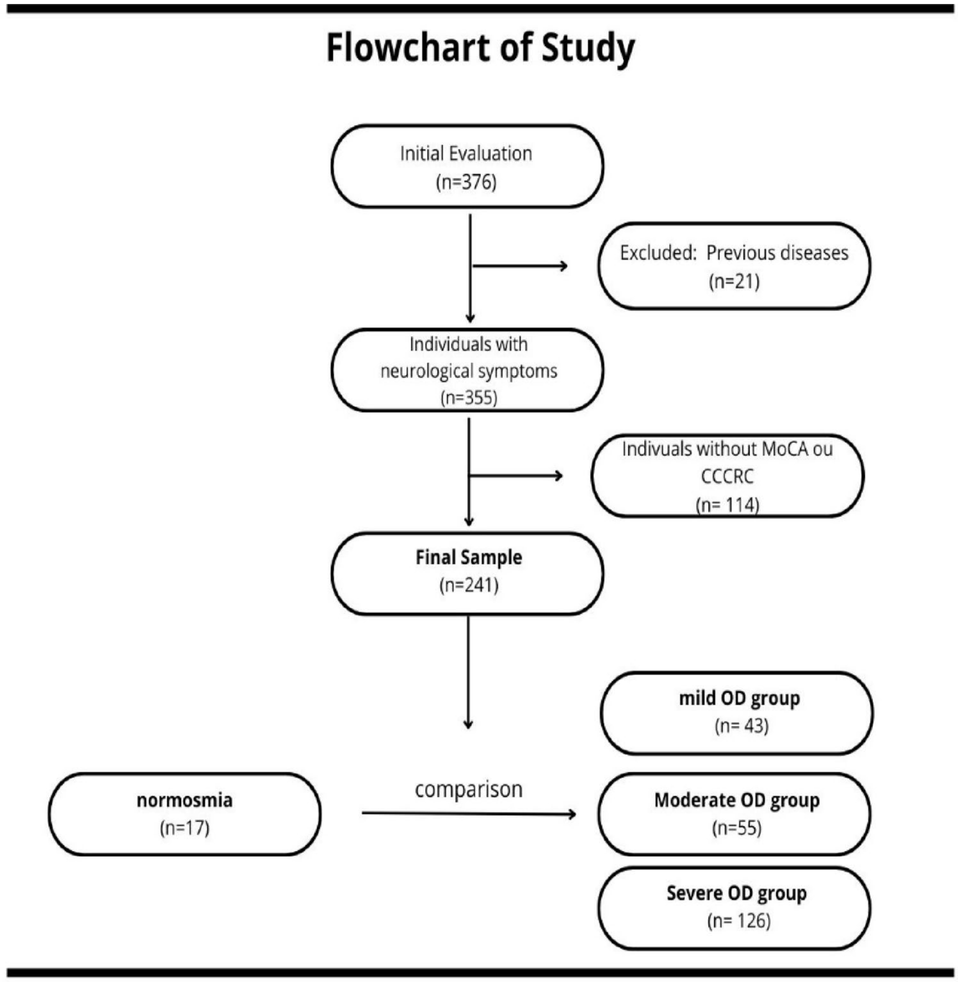

Fig. 1 Flowchart of Study

registered in electronic medical records (Microsoft Access; Microsoft Corp). The Montreal Cognitive Assessment scale, Brazilian version 8.1 (MoCA-BR) [41, 42] and the Connecticut Chemosensory Clinical Research Center (CCCRC) test [43, 44] were used to assess current levels of cognition and olfaction, respectively.

The MoCA is a brief screening tool for mild cognitive impairment, comprising a series of tasks that assess 6 cognitive domains: attention, memory, language, visuospatial abilities, executive functions, and spatial and temporal orientations. The MoCA takes approximately 10 min to administer, with a total score of 30 points; a score of 26 or higher is considered normal [41], but a cutoff score of 23 was used to decrease the false positive rate and for better accuracy, as recommended in recent studies [51, 52].

The CCCRC olfactory test comprises 2 subtests: a threshold test and an odour identification test. For the threshold test, a kit comprising 8 tubes was used. Seven tubes contained various dilutions of butanol (n-butyl alcohol) in distilled water, with butanol concentrations of 0.005%, 0.01%, 0.05%, 0.1%, 0.4%, 1%, and 4%, and 1 tube contained only distilled water. The procedure involved 2 alternatives with a forced choice between the butanol concentration and distilled water. The concentration of the first solution with 4 correct identifications was considered the perceptual threshold for detecting the odour of butanol [43, 44].

The odour identification test involved identifying 8 common substances by inhalation from a list of 16 items. The CCCRC score for each nostril was calculated by summing the scores from the threshold and odour identification tests, with scores ranging from 0 to 7. The final score was obtained by averaging the scores for both nostrils. Participants were then classified as having anosmia (0-1.75), severe hyposmia (2-3.75), moderate hyposmia (4-4.75), mild hyposmia (5-5.75), or normosmia (6–7) [43, 44].

The participants were divided into 4 groups: a group comprising individuals with normosmia, which was compared with other groups; a group consisting of individuals with mild hyposmia (mild OD group); a group with moderate hyposmia (moderate OD group); and a group with severe hyposmia or anosmia (severe OD group).

Statistical analysis

The collected data were organised into a database created using Excel 365 (Microsoft Inc.), and statistical analyses were performed using the StatPlus statistical software (AnalystSoft Inc.). Continuous numerical variables were expressed as means and standard deviations. After undergoing a normality test, variables with a parametric distribution were analysed using the 2-sample t-test, while non-parametric variables were analysed using the Mann-Whitney test. Age was analysed using one-way ANOVA with the Tukey post hoc test. Categorical variables were expressed as absolute and relative frequencies and analysed using the chi-square test. Binary logistic regression was conducted with cognitive impairment as the dependent variable and OD, number of related symptoms, and age as the independent variables. The odds ratio test was used to assess the association between OD and cognitive impairment. Pearson r-test was conducted for correlation analysis between the CCCRC and MoCA scores. To analyse the score of MoCA cognitive domains between groups, a one-way ANOVA test was conducted. An alpha level of 5% was used to reject the null hypothesis.

Results

The final study sample comprised 241 individuals with neurological complaints related to long COVID, predominantly women (179, 74%), with a mean age of 48.60 ± 12.68 years. Lower hospitalisation (40, 16%). The majority had up to 12 years of education (138, 57%) and reported a monthly income of at least US$470.00 (196, 81%). Cognitive impairment was the most frequently reported persistent symptom in the overall sample ($n = 154$, 64%), with 11 individuals in the normosmia group, 27 individuals (65%) in the mild hyposmia group, and 53 individuals (95%) in the moderate OD group. OD was the most frequently reported symptom in the severe OD group ($n = 85$; 67%). (Table 1).

Regarding the olfactory test results, 126 participants (52%) exhibited severe hyposmia to anosmia, 55 (22.8%) had moderate hyposmia, 43 (18%) had mild hyposmia, and 17 (7%) had normosmia. The results of the cognitive screening also indicated that many of the participants exhibited impairments, with 125 individuals (52%) showing scores suggestive of cognitive impairment (MoCA < 23); of these, 6 (35%) were in the normosmia group, 21 (48%) in the mild OD group, 20 (36%) in the moderate OD group, and 77 (51%) in the severe OD group.

Group comparison showed that the participants in the severe OD group were older than those in the normosmia group, without a statistically significant difference. Cognitive impairment was more frequently reported in the moderate OD group. Cognitive evaluation showed the worst results between the severe OD and has worse cognition than the normosmia group [22(5) vs. 25(3), $p < 0.05$], and the number of participants with cognitive impairment (MoCA < 23) was higher in the severe OD group ($n = 77$; 61%) (Table 1).

A binary logistic regression with cognitive impairment as the dependent variable and with age (40 years old or more), number of symptoms (more than 3), OD, and severe OD as independent variables was performed (Table 2).

The odds ratio of 2.19 and a 95% CI of (1.31–3.67), $p = 0.003$ indicates a higher likelihood that the group with severe OD will have cognitive impairment. Age was associated with a higher likelihood of cognitive impairment, OR 3.31 and 95% CI [1.72–6.37], $p = 0.000$ with cognitive impairment more likely in participants older than 40 years.

When conducting a correlation analysis between the total MoCA and CCCRC scores within the same population, an r of 0.14 ($p = 0.02$) was obtained, indicating a direct but weak correlation between the 2 variables.

To better understand the effect of each MoCA subscale on the reduction of MoCA scores, these domains were compared between study groups, and showed variation between groups in the visuospatial and executive domain (score 5), with a median (IQR) subscore in the moderate OD group of 2 (0), $p = 0.05$ (Table 3). Individuals with OD and cognitive impairment showed a reduction in all MoCA domains compared to individuals with OD without cognitive impairment and a statistically significant difference in the visuospatial and executive domain (3 [2] vs. 5 [1], $p = 0.00$) (Table 4).

Discussion

Persistent OD has been associated with decreased cognitive performance in the MoCA among 241 patients reporting neurological symptoms related to long COVID, particularly in women, adults, and those who were not hospitalised during the acute phase of COVID-19. Previous studies found similar results concerning the epidemiological profile associated with female sex, adult age, and lower hospitalisation rates in populations developing long COVID [45–47].

An odds ratio of 2.19 indicates an increased probability of cognitive impairment in those with severe OD compared with those without OD or with moderate to mild OD. The correlation between the total MoCA and CCCRC test scores was $r = 0.14$ ($p = 0.02$), indicating a weak yet direct correlation between the 2 variables. This statistical significance suggests that, while the degree of association may not be strong, there is a relevant connection between olfactory impairment and cognitive performance. This weak

Table 1 Sample characteristics

	General (n = 241)	Normosmia group (n = 17)	Mild OD group (n = 43)	Moderate OD group (n = 55)	Severe OD Group (n = 126)	p-value	p-value2	p-value3
Sex								
Women $_{n(\%)}$	179 (74%)	15 (89%)	33 (77%)	40 (72%)	91 (72%)	0.31	0.18	0.15
Men $_{n(\%)}$	62 (24%)	2 (11%)	10 (23%)	15 (28%)	35 (28%)			
Age $_{mean\pm sd}$	48,60 ± 12,6	43,94 ± 11,6	46,04 ± 12,2	47,92 ± 12,2	50.31 ± 12.9	0.56	0.24	0.06
Years of schooling								
Up to 12 years $_{n(\%)}$	138 (57%)	12 (70%)	21 (49%)	26 (47%)	79 (62%)	0.12	0.09	0.52
More than 12 years $_{n(\%)}$	103 (43%)	5 (30%)	22 (51%)	29 (53%)	47 (38%)			
Income								
Up to US$ 470.00** $_{n(\%)}$	196 (81%)	15 (89%)	25 (58%)	43 (78%)	109 (86%)	0.02#	0.36	0.84
More than US$ 470,00** $_{n(\%)}$	45 (19%)	2 (11%)	18 (42%)	12 (22%)	17 (14%)			
Hospitalization	40 (16%)	0	8 (18%)	12 (22%)	20 (16%)	0.05#	0.03#	0.07
Time of Long COVID $_{mean\pm sd}$								
Up to 90 days $_{n(\%)}$	12 (5%)	1 (6%)	4 (9%)	2 (3%)	5 (4%)	0.66	0.68	0.71
More than 90 days $_{n(\%)}$	230 (95%)	16 (94%)	39 (90%)	53 (97%)	121 (96%)			
Reported symptoms								
Cognitive impairment $_{n(\%)}$	154 (64%)	11 (65%)	27 (63%)	53 (96%)	62 (49%)	0.89	0.00#	0.23
Olfactory dysfunction $_{n(\%)}$	134 (55%)	5 (29%)	17 (39%)	27 (49%)	85 (67%)	0.46	0.15	0.002#
Headache $_{n(\%)}$	133 (55%)	13 (76%)	24 (56%)	37 (67%)	58 (46%)	0.13	0.47	0.01#
Anxyous symptoms $_{n(\%)}$	116 (48%)	12 (70%)	26 (60%)	35 (63%)	50 (39%)	0.46	0.59	0.01#
Gustatory dysfunctions $_{n(\%)}$	107 (44%)	6 (35%)	18 (42%)	21 (38%)	62 (49%)	0.64	0.83	0.28
Fatigue $_{n(\%)}$	103 (42%)	9 (53%)	20 (46%)	25 (45%)	48 (38%)	0.65	0.58	0.24
Dyspnoea $_{n(\%)}$	68 (28%)	5 (29%)	12 (28%)	14 (25%)	36 (28%)	0.90	0.74	0.94
Long COVID reported symptoms								
Up to 3 symptoms	136 (56%)	8 (48%)	21 (49%)	31 (56%)	77 (61%)	0.90	0.50	0.26
More than 3 symptoms	105 (44%)	9 (52%)	22 (51%)	24 (43%)	49 (39%)			
MoCA $_{\textbf{median (IQR)}}$	23(6)	25 (3)	23.5(5.75)	25(5.5)	22(5)	0.81	0.90	0.05##
MoCA ≤ 23	125 (52%)	6 (35%)	21 (48%)	20 (36%)	77 (61%)	0.34	0.93	0.04#
MoCA > 23 $_{n(\%)}$	116 (48%)	11 (64%)	22 (51%)	35 (64%)	49 (39%)			

OD: Olfactory dysfunction. p-value1: normosmia group vs. mild OD group. p-value2: normosmia group vs. moderate OD group. p-value3: normosmia group vs. Severe OD group. Long COVID reported symptoms such as gustatory dysfunction, olfactory dysfunction, Anxyous symptoms, Headache, Dyspnoea, Fatigue, Cognitive impairment. # Fisher's exact test (p < 0.05) ## Mann-Whitney U test (p < 0.05). **US$ 470.00 is equivalent of two Brazil's minimum wage

Table 2 Binary logistic regression, cognitive impairment as dependent variable

	OR	95%CI	Beta error	p-value
Moderate to Severe Olfactory Dysfunction (CCCRC score < 5.0)	1.32	(0.73–2.37)	0.27	0.35
Severe Olfactory Dysfunction (CCCRC score < 4.0)	2.19	(1.31–3.67)	0.78	0.003*
> 3 symptoms	1.27	(0.76–2.11)	0.24	0.35
40 years-old or more	3.31	(1.72–6.37)	1.20	0.000*

OR: Odds Ratio. 95% CI: 95% Confidence interval

correlation may reflect that OD is associated with cognition, and the association between these 2 symptoms is likely influenced by age, sex, mood disorders, and pre-existing comorbidities.

One review systematised the findings of 17 studies that investigated the relationship between cognition and olfaction in middle-aged adults following COVID-19. Significant associations between OD and cognitive impairment were found in 12 of the 17 studies, consistent with previous researches in the field of neurodegenerative diseases [48–50].

The MoCA screening test was employed in 5 [23, 24, 29, 39, 47] of the 17 studies, and in 3 [23, 29, 47], the results indicated a decrease in MoCA scores among participants with OD. Two of the 5 studies that used the MoCA assessed olfaction using standardised tests, the brief smell identification test (BSIT) and the University of Pennsylvania smell identification test (UPSIT), and found correlations between these variables [29, 47]. The authors highlighted that self-reporting as a method of assessment resulted in more heterogeneous and inconclusive outcomes, suggesting the need to evaluate individuals with persistent OD following COVID-19 through standardised and comprehensive tests alongside cognitive assessments to better understand the

Table 3 Comparison of MoCA domains between groups, considering MoCA score ≤ 23

	Normosmia group (n = 17)	Mild OD group (n = 43)	Moderate OD group (n = 55)	Severe OD Group (n = 126)	p-value
Visuospatial/Executive (Score 5)	3(1)	3(2)	**2(0)**	3(2)	0.05*
Naming (Score 3)	3(0.5)	3(1)	2.5(1)	3(1)	0.6
Attention (Score 6)	4(1)	3(2)	4(2)	4(2)	0.23
Language (Score 3)	3(0.5)	2(0.25)	2(0.75)	2(2)	0.43
Abstraction (Score 2)	2(0.5)	2(1)	1(1)	1(1)	0.79
Delayed recall (Score5)	0(0.5)	2(1.25)	2(1)	2(3)	0.09
Orientation (Score 6)	6(0)	6(0)	6(0.75)	6(1)	0.71

Data are presented as the median (IQR). * One-way ANOVA (p < 0.05). Cognitive domains evaluated by MoCA task: Visuospatial/Executive: visuospatial. abilities and executive function. Naming: memory. Attention: attention. Language: language. Abstraction: language. Delayed recall: memory. Orientation: spatial and temporal orientations

Table 4 Comparison of MoCA domains in individuals with olfactory dysfunction, considering MoCA score

MoCA domains	MoCA ≤ 23 (n = 118)	MoCA > 23 (n = 106)	p-value
Visuospatial/ Executive (Score 5)	3(2)	5(1)	0.000*
Naming (Score 3)	3(1)	3(0)	0.002*
Attention (Score 6)	4(2)	5(1)	0.000*
Language (Score 3)	2(2)	3(1)	0.000*
Abstraction (Score 2)	2(1)	2(0)	0.000*
Delayed recall (Score 5)	2(2)	3(2)	0.000*
Orientation (Score 6)	6(1)	6(0)	0.007*

Data are presented as medians (IQR). Mann-Whitney test (p < 0.05). Cognitive domains evaluated by MoCA task: Visuospatial/Executive: visuospatial. abilities and executive function. Naming: memory. Attention: attention. Language: language. Abstraction: language. Delayed recall: memory. Orientation: spatial and temporal orientations

impact of long COVID in these patients [48]. Executive function is an important domain of cognition that is affected by long COVID and can substantially impact the ability of individuals to perform daily living activities [53].

This study had some limitations that should be considered when interpreting the findings. The sample size, particularly in the normosmia group, may limit the generalisation of the results and reduce the statistical power to detect significant associations. The lack of neuroimaging examinations in this cohort does not permit the identification of affected areas in the brain. Indeed, our findings, along with those of previous studies, underscore the need for further investigation into the underlying mechanisms linking OD to cognitive impairment in patients with long COVID [34, 49].

Conclusion

This study suggests an association between persistent OD and cognitive impairment in patients with long COVID. This supports a growing body of evidence indicating that COVID-19 may lead to persistent neurological effects, highlighting the importance of monitoring the cognitive and olfactory functions in these patients. Early detection of OD could serve as a warning sign for a more rigorous cognitive follow-up, which is relevant for developing therapeutic and rehabilitation strategies.

Additional longitudinal and prospective studies with large sample sizes recruited from the general population and multicentre studies are necessary to generalise the results and establish a causal or temporal relationship between olfactory and cognitive impairment, how olfaction provides insight into the central nervous system in individuals affected by chronic forms of COVID-19, and how this correlation is similar to or diverges from the mechanisms explored in research involving OD and neurodegenerative diseases. The inclusion of neuroimaging examinations in future studies will improve the understanding of the neurological mechanisms and brain areas affected.

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Data availability The data underlying this article will be shared on reasonable request to the corresponding author.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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